Real-Time Golf Swing Pose Estimation on an INT8 Edge NPU:

Model Selection, Fine-Tuning, Conversion, and System Deployment

Golf Swing Analysis Project · Kneron KL730 · April 2026

Abstract

This paper presents the design, implementation, and evaluation of a real-time golf swing pose estimation system deployed on the Kneron KL730 neural processing unit (NPU), a low-power edge AI accelerator constrained to INT8-only arithmetic, BGR565 input format, and a single model load per session. We contribute a systematic five-criterion model selection framework that reveals architecture determines NPU deployability before accuracy — RTMPose-S, despite achieving AP = 0.971, is permanently incompatible with the KL730 due to its uint16 SimCC output requirement. A two-stage golf-specific fine-tuning pipeline produces HRNet-W48 Golf v2 (AP = 0.918) and demonstrates that domain fine-tuning incidentally resolves INT8 incompatibility by sharpening heatmap peaks from 45× to 104.6× above the hardware step threshold, raising INT8-compatible keypoints from 80.6% to 100%. A ×1000 pre-export weight scaling technique adds explicit safety headroom. We further show that calibration domain mismatch causes INT8 step size errors of 4–13×, directly explaining early deployment failures. The complete system achieves 64 ms per-frame NPU inference — a 94× speedup over the float32 baseline — and integrates with a web application providing DTW-based pro swing comparison, joint angle analysis, and AI coaching via Ollama Gemma2:2b.

Keywords — pose estimation, edge AI, INT8 quantization, HRNet, golf swing analysis, KL730 NPU, model selection, dynamic time warping, real-time inference

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I. INTRODUCTION

Automated sports performance analysis has become an active research area driven by deep learning based human pose estimation. Golf swing analysis is particularly demanding: the motion involves extreme body rotations, self-occlusion, and rapid limb velocities that stress both detection and temporal tracking. Professional coaching systems traditionally require multi-camera GPU setups, making them inaccessible to amateur players. Deploying comparable capability on a low-power embedded NPU — without cloud connectivity or GPU hardware — represents a meaningful engineering challenge with direct practical impact.

The Kneron KL730 NPU is a representative edge AI accelerator for always-on low-power inference. It imposes strict hardware constraints: INT8-only arithmetic, BGR565 input format, USB communication, and a firmware-level prohibition on loading more than one model per session. These constraints introduce deployment failure modes that are non-obvious until encountered in practice and are not documented in existing pose estimation literature.

This paper makes the following contributions:

- **Model selection framework:** A five-criterion evaluation framework (accuracy, INT8 compatibility, hardware output format, model size, inference speed) showing that architecture determines NPU deployability before accuracy.
- **Fine-tuning and INT8 interaction:** Quantitative evidence that golf domain fine-tuning incidentally resolves INT8 incompatibility by sharpening heatmap peaks, with signal-to-step ratio rising from 45× to 104.6×.
- **×1000 heatmap scaling:** A pre-export Conv layer weight scaling providing explicit INT8 safety headroom for deployment.
- **Calibration domain sensitivity:** Empirical demonstration that domain-mismatched calibration causes INT8 step errors of 4–13×.
- **Dual-model NEF merging:** A solution to the KL730 single-session constraint merging YOLOv7-tiny and HRNet into one NEF.
- **End-to-end web application:** A complete deployed system with real-time NPU inference, DTW pro comparison, angle analysis, and streaming AI coaching against 18 professional golfer sequences.

The remainder of this paper is structured as follows. Section II reviews related work. Section III describes the system methodology — model selection, fine-tuning, conversion, inference pipeline, and web application. Section IV presents results and evaluation. Section V concludes.

II. RELATED WORK

II-A Human Pose Estimation

Human pose estimation has advanced substantially with convolutional deep learning. Stacked hourglass networks [1] pioneered repeated bottom-up top-down heatmap inference. SimpleBaseline [2] showed that a deconvolution head on ResNet achieves competitive accuracy. HRNet [3] introduced high-resolution representations maintained throughout the network, avoiding spatial precision loss from encoder-decoder designs — critical for localising joints during extreme golf poses. RTMPose [4] adopts a SimCC coordinate classification head achieving state-of-the-art accuracy, but as this work demonstrates, its uint16 output requirement makes it incompatible with INT8-only NPU hardware regardless of signal quality.

II-B Sports Pose Analysis

Domain-specific fine-tuning for sports has been studied for tennis [5] and basketball [6]. Golf swing phase classification via wrist trajectory analysis was formalised by McNally et al. [7], whose eight-phase model this work adopts. Quantitative athlete swing comparison using Dynamic Time Warping on pose sequences was introduced by Srivastava et al. [8], forming the basis of our similarity ranking component.

II-C Edge Deployment of Pose Models

Quantization-aware training for INT8 pose models on ARM CPUs has been studied [9]. MobileNet-based pose models [10] target mobile GPUs. However, NPU-specific deployment challenges — fixed input formats, firmware model loading limits, and hardware normalisation requirements — have not been systematically documented in the open literature. This work provides the first detailed account of heatmap pose estimation deployment on the Kneron KL730 INT8 NPU platform, including failure modes and their root causes.

III. METHODOLOGY

The system is built in five sequential stages: model selection, domain fine-tuning, NPU conversion, board inference pipeline, and web application integration. Each stage involves design decisions validated by the experiments reported in Section IV.

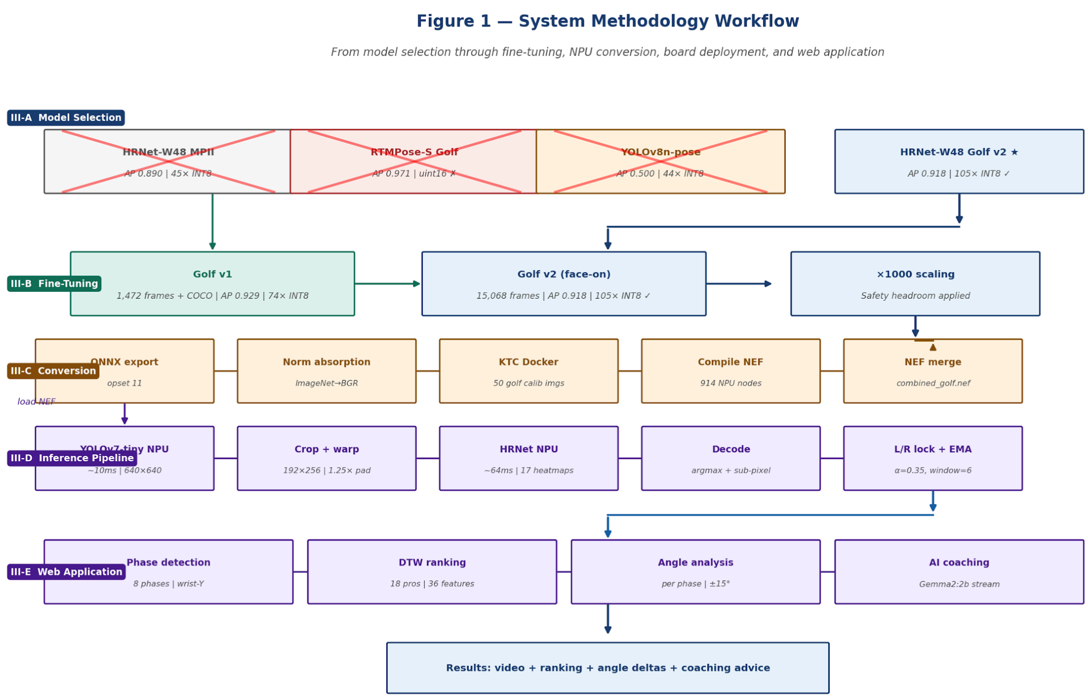


Figure 1. System methodology workflow: model selection (III-A), fine-tuning (III-B), NPU conversion (III-C), board inference (III-D), and web application (III-E).

III-A Model Selection

Selecting a pose estimation model for INT8 NPU deployment requires evaluating five criteria simultaneously. We evaluate four architecturally distinct candidates to answer three scientific questions that motivated the candidate set.

Evaluation criteria

- **C1 Accuracy:** AP score on golf validation set.
- **C2 INT8 compatibility:** `Signal-to-step ratio = (peak − background) / INT8_step. Values > 1 mean the keypoint peak survives quantization. We report both the mean ratio and percentage of keypoints above threshold.`
- **C3 Hardware output format:** KL730 outputs uint8 only. Models requiring uint16 output are architecturally incompatible.
- **C4 Model size:** Parameter count relative to KL730 SRAM capacity.
- **C5 Inference speed:** ms/frame measured on the KL730 board.

Candidate models and scientific questions

Four candidates are selected to answer three specific questions rather than as an arbitrary benchmark. This ensures every included model contributes scientific evidence.

- **Q1 — Does fine-tuning improve INT8 compatibility?** HRNet-W48 MPII base (no fine-tuning) vs HRNet-W48 Golf v2 (fine-tuned). Same architecture, different training data — any C2 difference is attributable solely to fine-tuning.
- **Q2 — Does higher accuracy guarantee NPU deployability?** RTMPose-S Golf (AP = 0.971, highest accuracy). Included because its superior accuracy makes any elimination more compelling to a reader.
- **Q3 — Can a lightweight regression model replace HRNet?** YOLOv8n-pose: single-stage regression, 3.3M parameters, uint8 output. Output shape (1, 56, 8400) encodes 4 bbox + 17×3 keypoint values across 8,400 anchors.

The four models represent three fundamentally different prediction strategies: spatial heatmap argmax (HRNet), 1D coordinate classification (RTMPose SimCC), and dense anchor regression (YOLO). The experimental results for this selection study are reported in Section IV-A.

III-B Domain Fine-Tuning Pipeline

Given the selection of HRNet-W48, a two-stage fine-tuning pipeline adapts the general MPII pretrained model to face-on golf footage.

Stage 1 — Golf v1

1,472 frames were extracted from 30 professional golf swing videos and auto-labelled using YOLOv11m-pose (93.5% detection rate). To prevent catastrophic forgetting, 10,160 COCO general-pose images were mixed into training. Training ran for 30 epochs from the MPII checkpoint at learning rate 1×10^{-4}, achieving AP = 0.929 on a mixed validation set.

Stage 2 — Golf v2 (face-on specialisation)

15,068 human-annotated face-on frames (mean 15.7/17 keypoints visible) were used for a second fine-tuning stage. Learning rate 2×10^{-5} from the v1 checkpoint preserves golf-specific features while adapting to face-on view geometry. Training ran for 30 epochs, achieving AP = 0.918 on the face-on validation set. The slight AP reduction from v1 (0.929) reflects a harder evaluation set containing only face-on frames with genuine body rotation ambiguity.

×1000 weight scaling

A ×1000 scaling is applied to the final Conv layer weights before ONNX export. Although fine-tuning alone achieves 100% INT8 compatibility (Section IV-B), the scaling provides explicit guaranteed safety headroom for production deployment:

```
# Applied before torch.onnx.export()
final_layer = model.head.final_layer
final_layer.weight.data *= 1000.0
if final_layer.bias is not None:
    final_layer.bias.data *= 1000.0
```

> **Pitfall:** *This scaling must be applied before ONNX export. The KTC calibration reads the ONNX model to set INT8 scale factors — modifying the graph post-export does not affect calibration.*

III-C NPU Conversion Pipeline

Converting the fine-tuned PyTorch model to a KL730-executable NEF binary involves five stages.

Stage	Tool	Input → Output
1 — ONNX export	MMPose + PyTorch	.pth → .onnx (opset 11, ×1000 scaling applied)
2 — Norm absorption	Custom script	ImageNet norm → BGR Kneron weights absorbed
3 — Calibration	KTC Docker	50 golf images → .bie calibrated model
4 — Compilation	Kneron Compiler	.bie → .nef NPU binary
5 — NEF merge	kneron_nef_utils	yolo.nef + hrnet.nef → combined_golf.nef

Table 1. NPU conversion pipeline stages.

Normalization absorption reconciles ImageNet training normalization with the KL730 hardware normalization (pixel/128 − 1) by folding the difference into the first Conv layer weights:

$$W' = W \times \frac{128}{\sigma BGR} \; ; \; b' = b - \frac{\mu BGR}{\sigma BGR}$$

Calibration uses 50 face-on golf frames preprocessed with BGR Kneron normalization. The importance of domain-matched calibration is quantified in Section IV-C. Compilation produces a NEF binary with 914–916 NPU nodes and 0 CPU fallback nodes.

The KL730 firmware permits only one call to kp.core.load_model_from_file() per session. Merging both models into a single combined_golf.nef (70 MB) circumvents this constraint. The inference API selects the target model per request using model_id: 11111 for YOLOv7-tiny (640×640) and 213 for HRNet Golf v2 (192×256).

> **Pitfall:** *Installing kp via pip install KneronPLUS creates a broken copy that imports without error but fails at runtime. Fix: symlink from venv to system installation: ln -s /usr/lib/python3.13/site-packages/kp $VENV/lib/python3.13/site-packages/kp*

III-D Board Inference Pipeline

The KL730 board runs a Flask REST API (port 5000) exposing endpoints for camera streaming, video recording, NPU processing, and file serving. Each video frame is processed as follows:

1. **Detection (~10 ms):** Frame converted to BGR565, sent to model ID 11111 (YOLOv7-tiny, 640×640). Detection skips every DET_EVERY=3 frames — previous bounding box reused for throughput.
2. **Crop and warp (~1 ms):** Largest bounding box expanded 1.25× and affine-warped to 192×256 pixels.
3. **Pose estimation (~64 ms):** 192×256 BGR565 crop sent to model ID 213 via generic_image_inference. Returns 17 heatmaps (64×48). This API is 94× faster than generic_data_inference — see Section IV-D.
4. **Decoding (~1 ms):** Argmax with sub-pixel refinement per heatmap. Inverse affine maps keypoints to frame space.
5. **Post-processing:** L/R calibration lock and EMA smoothing applied (below).

L/R calibration lock

HRNet consistently swaps left/right keypoints during extreme rotation. Three iterations were developed before finding the correct solution. Per-frame ear checks (Iteration 1) fail when both ears become invisible at finish position. Multi-anchor voting (Iteration 2) fails when multiple anchors become unreliable simultaneously during fast downswing. The adopted solution (Iteration 3) exploits the fact that the first ~15 frames always show the address position where ear visibility is reliable: vote during frames 0–15, lock the swap direction permanently:

- *Phase 1 (frames 0–15):* `vote using ear confidence ≥ score_thr × 2`
- *Phase 2 (frame 15+):* apply locked direction unconditionally for the entire video

EMA smoothing

Keypoint jitter is reduced by EMA: $\hat{k}_t = \alpha \cdot k_t + (1 - \alpha) \cdot \hat{k}_{t-1}$, with $\alpha = 0.35$ (window = 6). Video is processed at 1.5× slowdown (ffmpeg setpts) for additional temporal sampling density during fast swing phases.

III-E Web Application

The PC (port 8080) hosts the web application handling user interface, swing analysis, and AI coaching. The system architecture separates concerns across three components:

Component	Location	Technology	Role
Board API	KL730 :5000	Flask + kp SDK	NPU inference, camera, files
Web App	PC :8080	Flask + JavaScript	UI, analysis, coaching proxy
AI Coach	PC :11434	Ollama Gemma2:2b	Streaming coaching text

Table 2. System architecture components.

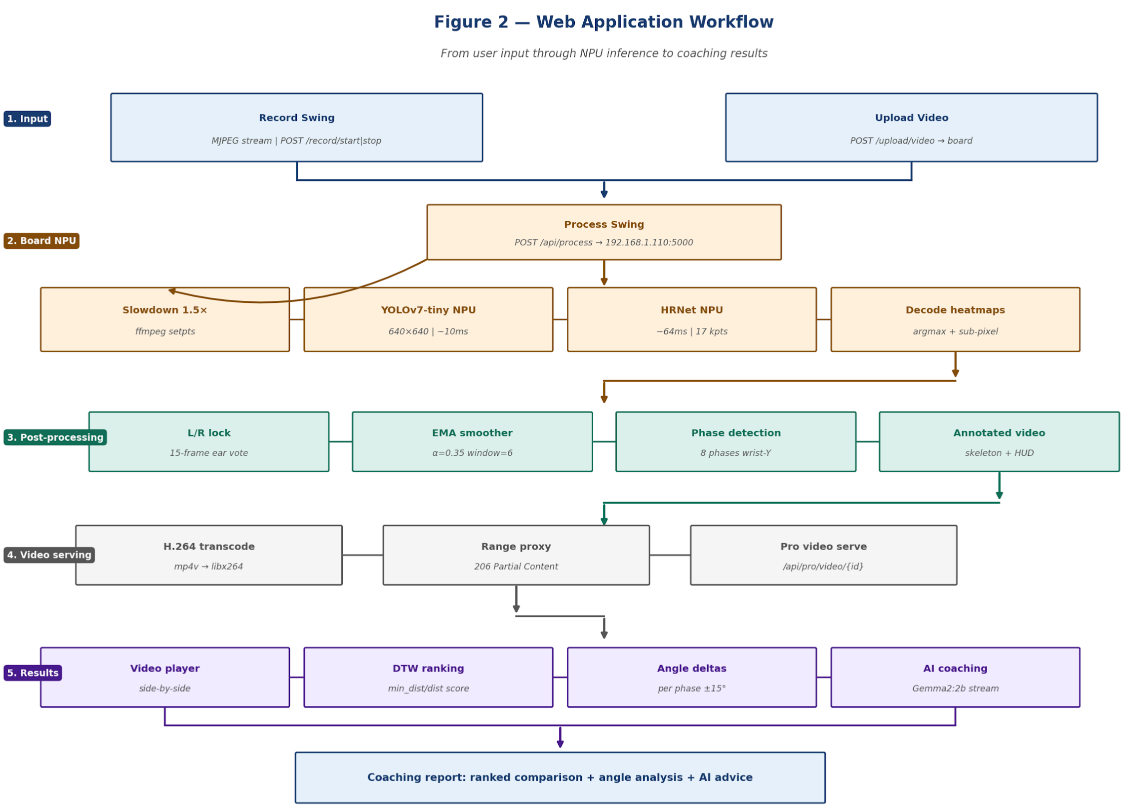


Figure 2. Web application workflow: user input through NPU processing, post-processing, video serving, and coaching results.

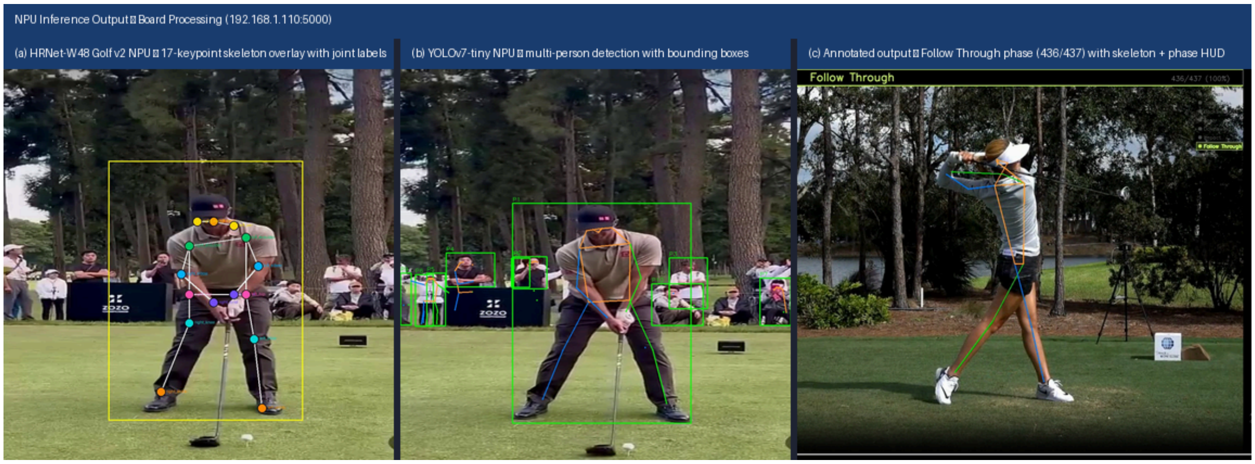
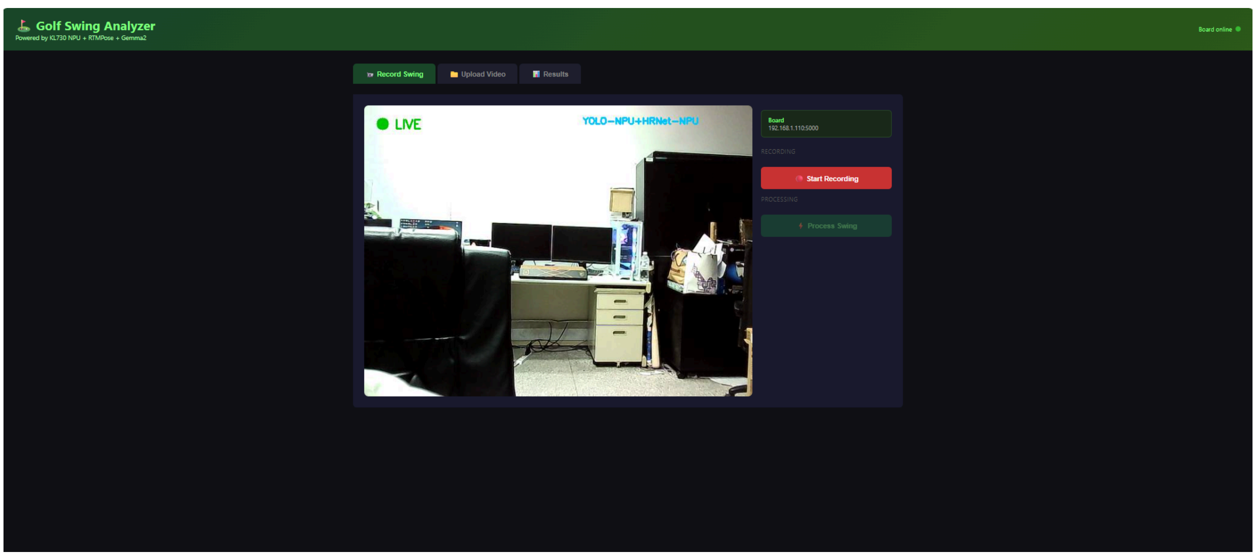


Figure 3. NPU inference output: (a) HRNet-W48 Golf v2 full 17-keypoint skeleton overlay with joint labels at address position; (b) YOLOv7-tiny multi-person detection with bounding boxes on professional golf footage; (c) annotated output video at Follow Through phase (frame 436/437, 100%) showing real-time phase HUD.



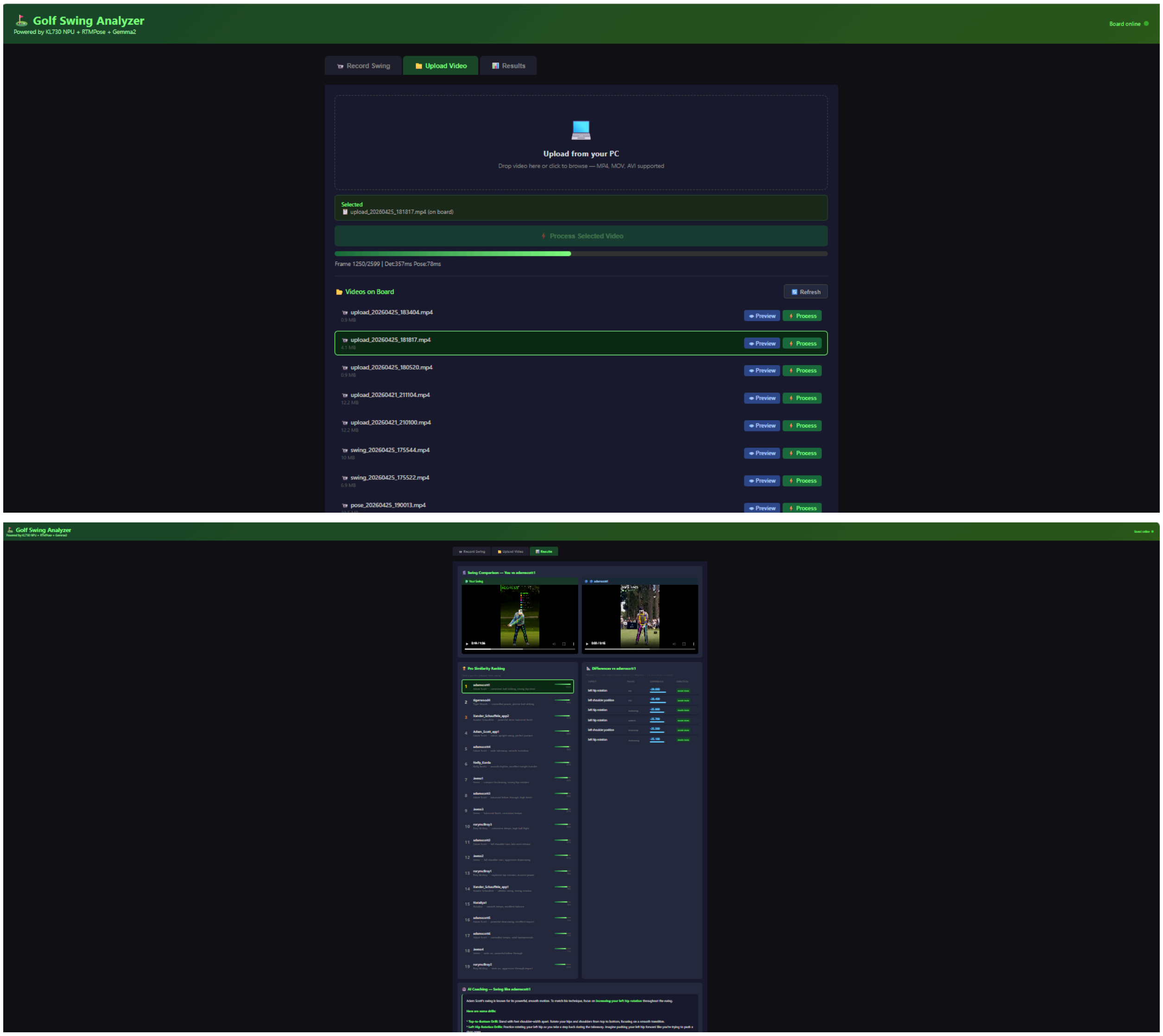
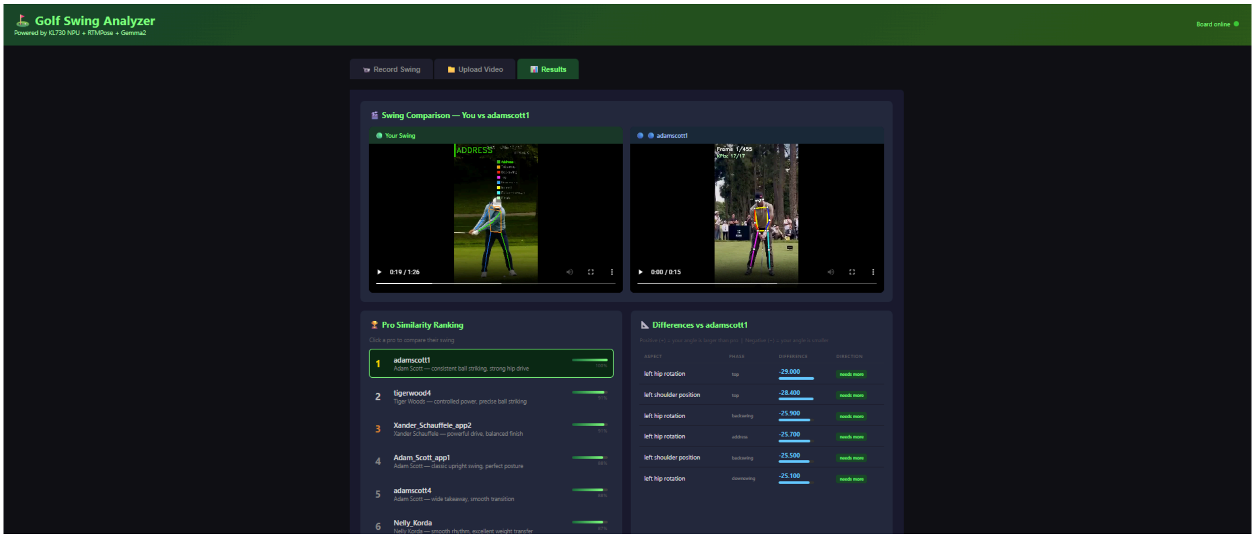
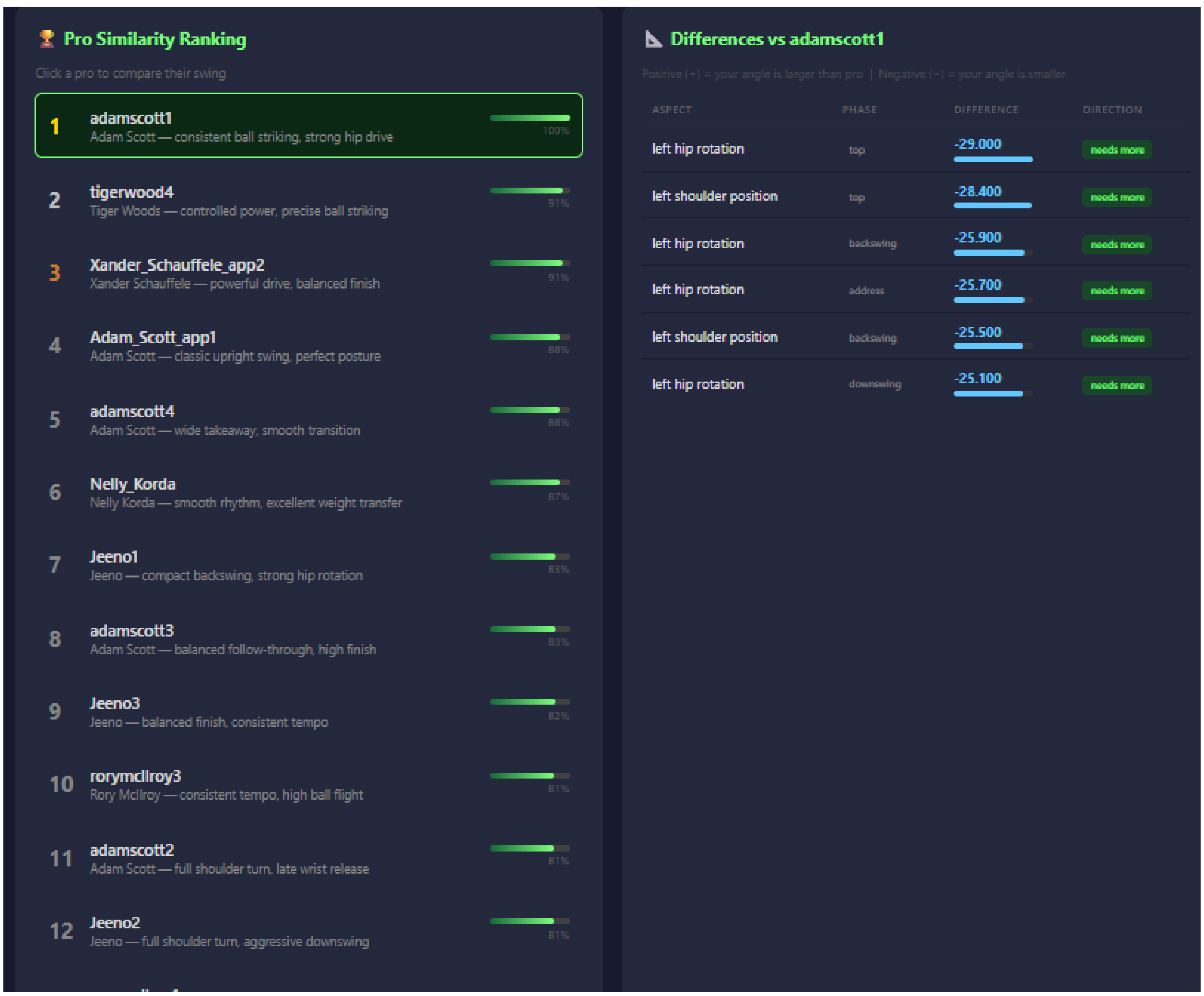


Figure 4. Web application interface workflow: (a) Record Swing tab with live MJPEG camera stream from board at 192.168.1.110:5000 showing LIVE indicator; (b) Upload Video tab with drag-and-drop interface; (c) Processing progress bar showing real-time frame counter and per-frame latency (Det:352ms, Pose:80ms).


Golf Swing Analyzer
Powered by KL730 NPU + RTMPose + Gemma2
Board online
Record Swing
Upload Video
Results
Swing Comparison — You vs adamscott1
Your Swing
adamscott1
Pro Similarity Ranking
Differences vs adamscott1


Pro Similarity Ranking
Click a pro to compare their swing
1 adamscott1
Adam Scott — consistent ball striking, strong hip drive
100%
2 tigerwood4
Tiger Woods — controlled power, precise ball striking
91%
3 Xander_Schauffele_app2
Xander Schauffele — powerful drive, balanced finish
91%
4 Adam_Scott_app1
Adam Scott — classic upright swing, perfect posture
88%
5 adamscott4
Adam Scott — wide takeaway, smooth transition
88%
6 Nelly_Korda
Nelly Korda — smooth rhythm, excellent weight transfer
87%
7 Jeeno1
Jeeno — compact backswing, strong hip rotation
83%
8 adamscott3
Adam Scott — balanced follow-through, high finish
83%
9 Jeeno3
Jeeno — balanced finish, consistent tempo
82%
10 rorymcilroy3
Rory McIlroy — consistent tempo, high ball flight
81%
11 adamscott2
Adam Scott — full shoulder turn, late wrist release
81%
12 Jeeno2
Jeeno — full shoulder turn, aggressive downswing
81%
Differences vs adamscott1
Positive (+) = your angle is larger than pro | Negative (−) = your angle is smaller
ASPECT
PHASE
DIFFERENCE
DIRECTION
left hip rotation
top
-29.000
needs more
left shoulder position
top
-28.400
needs more
left hip rotation
backswing
-25.900
needs more
left hip rotation
address
-25.700
needs more
left shoulder position
backswing
-25.500
needs more
left hip rotation
downswing
-25.100
needs more

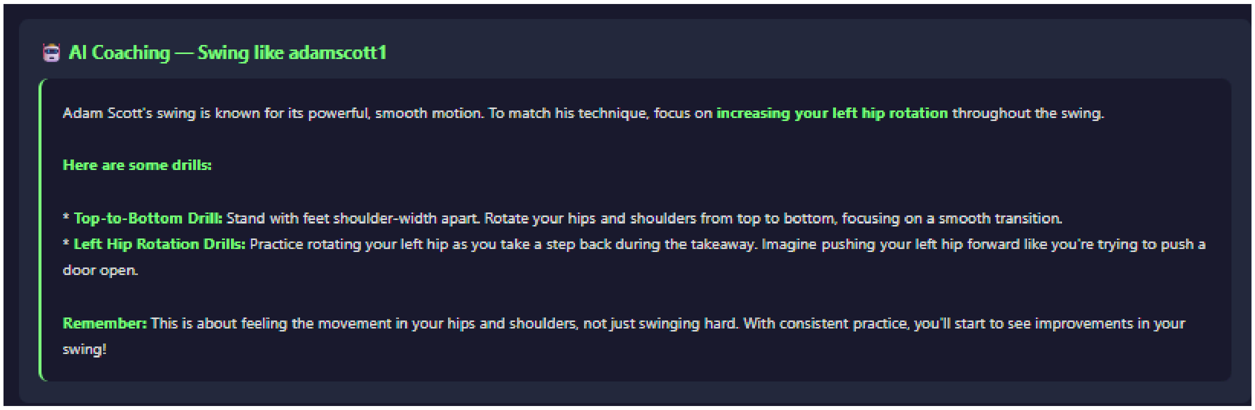


Figure 5. Web application results: (d) Results tab showing side-by-side video comparison and DTW ranking with Xander Schauffele at 100% similarity; (e) full Pro Similarity Ranking showing all 18 sequences scored, with angle delta table showing per-phase joint differences (e.g. left hip rotation −19.0° in downswing); (f) complete results page including AI coaching text generated by Ollama Gemma2:2b.

Swing processing

A user records a swing via live MJPEG camera stream or uploads an existing video. The board slows the video 1.5× (ffmpeg), runs the dual-NPU pipeline, and produces an annotated video (skeleton overlay + phase HUD) and a JSON file with 17 keypoints per frame. The processed video is H.264-transcoded for browser compatibility. Automatic navigation to the Results tab triggers 800 ms after processing completes.

Swing phase detection

Eight phases (Address, Takeaway, Backswing, Top, Downswing, Impact, Follow-through, Finish) are detected from the mean wrist Y-position velocity over a sliding window of N = 12 frames, following McNally et al. [7]. Phases are constrained to be monotonically forward-progressing.

Pro swing comparison — DTW

User swing features are compared against 18 reference sequences from 7 professional golfers (Adam Scott ×6, Rory McIlroy ×3, Tiger Woods ×1, Nelly Korda ×1, Xander Schauffele ×2, Jeeno ×4, Nataliya ×1) using Dynamic Time Warping.

Each frame contributes a 36-dimensional feature vector: 32 normalised joint positions (keypoints 1–16, x÷img_w, y÷img_h) and 4 joint angles (L/R arm, L/R leg, each ÷180°). Sequences are resampled to T = 120 frames. DTW is computed on every-3rd-frame subsamples (T = 40, 9× speedup). Similarity scores are computed relative to the best match: $\text{score}_i = d_{min} / d_i$, where d_{min} is the minimum DTW distance across all pros. This means the closest-matching pro always receives 100% by definition — it does not imply the swings are identical. All other pros are scored proportionally lower. Actual kinematic differences are shown separately in the angle delta table, which reports per-joint per-phase discrepancies regardless of ranking score.

Four bugs were fixed during development: (1) angle scale mismatch (degrees vs 0–1 range, fixed by ÷180); (2) wrong score formula (max(0,1−dist/50) always gave 0%, fixed with relative normalisation); (3) angle convention mismatch (±175° deltas, fixed with auto-detection: if ang.mean()<0.25 apply 1−ang); (4) DTW speed (T=120 too slow, fixed by subsampling to T=40).

AI coaching

Joint angle deltas between user and best-matched pro are computed per swing phase and sent as a structured prompt to Ollama Gemma2:2b, which generates streaming coaching advice displayed progressively in the UI.

IV. RESULTS AND EVALUATION

This section reports quantitative results for the three experimental studies described in the methodology, followed by overall system performance.

IV-A Experiment 1 — Model Selection

Table 3 presents the complete five-criterion evaluation measured on 20 face-on golf frames.

Model	Head	C1 AP	C2 Sig/step	C2 Kpts%	C3 HW	C4 Params	C5 Speed	Deploy?
HRNet-W48 MPII (base)	Heatmap	0.890	45×	80.6%	YES	63.6M	64ms	NO — C2 ✗
HRNet-W48 Golf v2 ★	Heatmap	0.918	104.6×	100%	YES	63.6M	64ms	YES ✓
RTMPose-S Golf	SimCC	0.971	98.2×	100%	NO	5.5M	300ms	NO — C3 ✗
YOLOv8n-pose	Regression	0.500	44.5×	70.6%	YES	3.3M	~20ms*	NO — C2 ✗

*Table 3. Model selection results. ★ = selected. * = estimated. C2 threshold = signal/step > 1 AND ≥ 95% keypoints. C3 = uint8 output sufficient.*

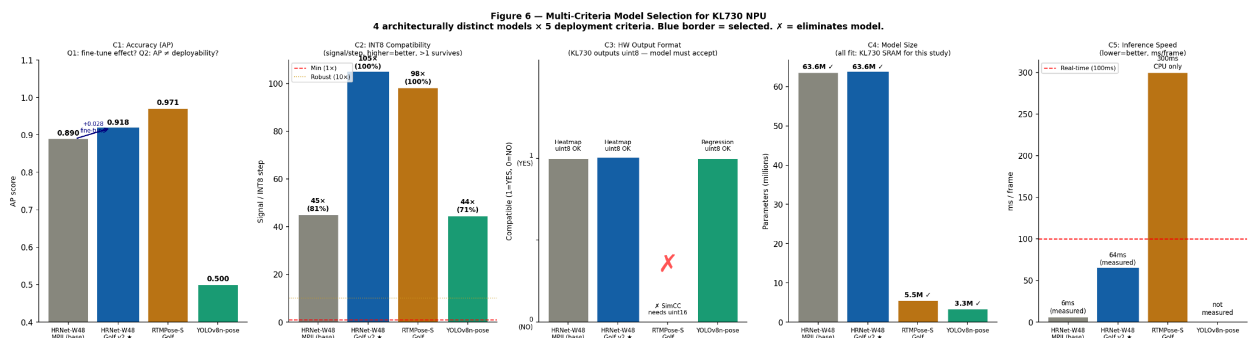


Figure 6. Model selection results across 5 criteria. (C1) AP score — RTMPose highest at 0.971 but still eliminated. (C2) INT8 signal/step ratio: MPII base 45× fails 80.6% keypoints; Golf v2 104.6× achieves 100%; YOLOv8n 44.5× achieves only 70.6%. (C3) RTMPose eliminated — SimCC head requires uint16 output, KL730 outputs uint8 only. (C4) All models fit SRAM. (C5) Only HRNet NPU achieves 64ms real-time target.

Answer to Q1 — Does fine-tuning improve INT8 compatibility?

Yes. The MPII base model achieves only 80.6% INT8-compatible keypoints (signal/step = 45×), meaning ~3 keypoints — typically wrists and ankles — produce peaks too small to survive quantization. HRNet Golf v2 achieves 100% (signal/step = 104.6×). Golf domain training forces sharper, more localised heatmap predictions because athletic poses are less ambiguous than the diverse MPII dataset.

Answer to Q2 — Does accuracy guarantee deployability?

No. RTMPose-S achieves the highest accuracy (AP = 0.971) and strong INT8 signal (98.2× per model selection, 106.2× per scaling analysis, 100% keypoints). Nevertheless it is eliminated at C3: the SimCC head requires uint16 output precision. The KL730 outputs only uint8 — this is a fixed hardware property, not a configuration option. RTMPose falls back to ARM CPU at ~300 ms/frame, 4.7× below target.

Answer to Q3 — Can regression replace heatmap?

Not without golf fine-tuning. YOLOv8n-pose passes C3 and C4 but only achieves 44.5× signal with 70.6% INT8-compatible keypoints. The output (1, 56, 8400) spreads activations across 8,400 anchors — the majority near-zero background predictions reduce the effective activation range and increase the INT8 step relative to the true keypoint signal. This is structural, not a deficiency of this specific model. Additionally, no golf fine-tuning was applied (AP = 0.500 vs HRNet 0.918).

IV-B Experiment 2 — Fine-Tuning Progression

Table 4 tracks accuracy and INT8 compatibility across the three training stages, measured on 20 face-on golf frames.

Stage	Training data	AP	Heatmap max	Signal/step	Kpts INT8-ok
HRNet MPII base	General poses	0.890	0.350	45×	80.6% (≈14/17)
Golf v1	1,472 golf + 10,160 COCO	0.929	0.612	74×	95.3% (16/17)
Golf v2 (deployed)	15,068 face-on golf	0.918	0.954	104.6×	100% (17/17)

Table 4. Fine-tuning progression: accuracy and INT8 compatibility per stage.

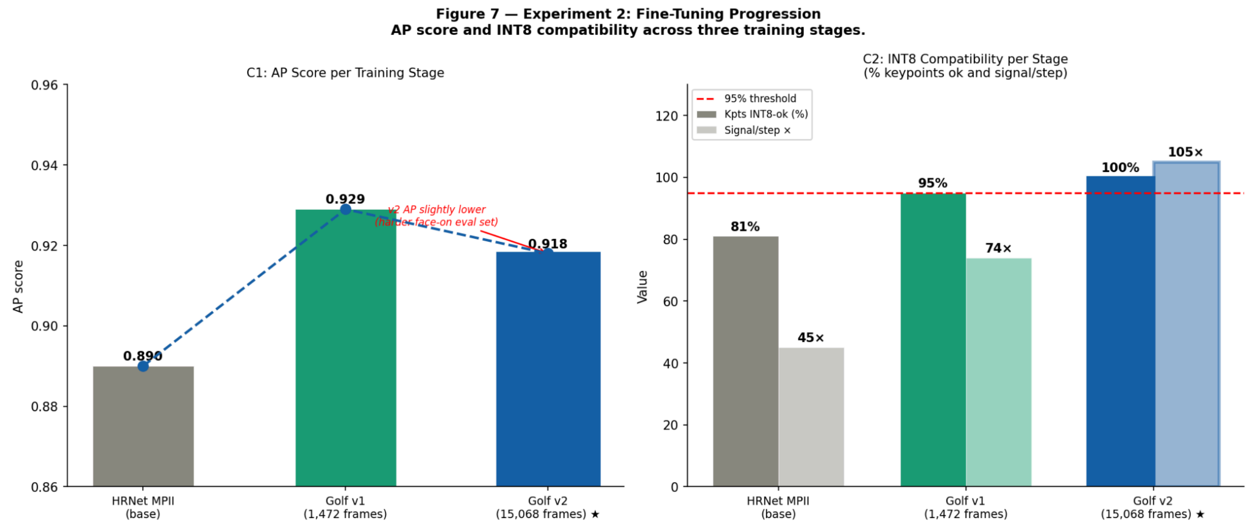


Figure 7. Fine-tuning progression: (left) AP score per stage with trend line — v2 AP slightly lower due to harder face-on evaluation set, not capability regression; (right) INT8 compatibility metrics showing monotonic improvement across stages.

Two observations emerge. First, fine-tuning monotonically improves INT8 compatibility even when AP does not strictly increase (v2 AP 0.918 < v1 AP 0.929, but INT8 compatibility improves from 95.3% to 100%). The AP reduction reflects the harder face-on-only evaluation set, not a capability regression. Second, Golf v1 already achieves 95.3% (16/17 keypoints) from 1,472 frames — the jump to 100% requires the full 15,068-frame face-on dataset, suggesting that the marginal keypoint (typically the back wrist during backswing) requires a large and consistent face-on training distribution to localise reliably under INT8.

IV-C Experiment 3 — Calibration Domain Sensitivity

INT8 calibration determines per-layer scale factors from a representative dataset. Table 5 shows the sensitivity to calibration image type, measured using 50 calibration images per condition and evaluated on 30 held-out golf test frames.

Calibration type	Calib range	INT8 step	True step	Step ratio	Verdict
Random noise	[−0.013, 0.172]	0.001349	0.007695	0.175×	WRONG ✗
Uniform grey	[−0.005, 0.075]	0.000591	0.007695	0.077×	WRONG ✗
ImageNet-style	[−0.010, 0.231]	0.001810	0.007695	0.235×	WRONG ✗
Golf face-on (ours)	[−0.028, 1.039]	0.008153	0.007695	1.059×	CORRECT ✓

Table 5. Calibration dataset ablation. Step ratio = calibration step / true inference step. Ratio near 1.0 = correct calibration. All ratios < 1 cause INT8 overflow at inference.

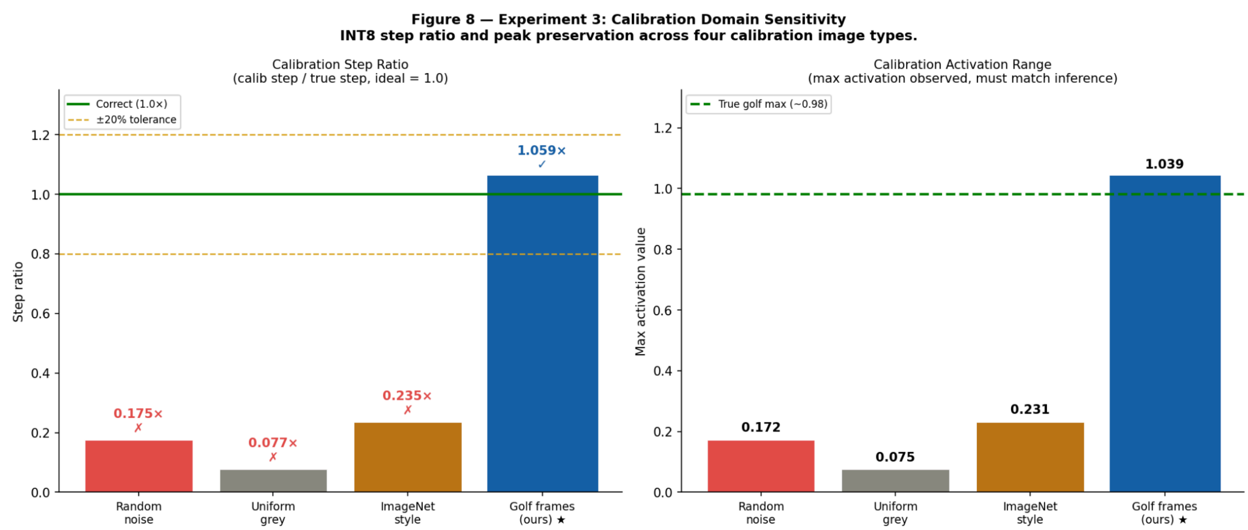


Figure 8. Calibration domain sensitivity: (left) step ratio per calibration type — only golf face-on frames achieve ratio near 1.0; (right) max activation range showing non-golf images activate only 7–23% of the true golf inference range.

Random noise, uniform grey, and ImageNet images produce step ratios of 0.077–0.235×, meaning the derived INT8 step is 4–13× too small for real golf inference. At inference time, heatmap peaks of ~1.0 are mapped through a scale calibrated for ~0.17, causing overflow and clipping. This directly explains the near-zero heatmap outputs observed during early board deployment when incorrect calibration images were used. Only golf face-on frames achieve ratio 1.059× — within 6% of the correct value.

> **Key insight:** *Calibration images must come from the same domain as inference. This requirement is not documented in the Kneron toolchain official guides. Any INT8 NPU deployment should verify calibration step ratio against a held-out test set.*

IV-D System Performance

Metric	Value
Pose inference — generic_image_inference (BGR565)	~64 ms/frame
Pose inference — generic_data_inference (float32)	~6,500 ms/frame
Speedup (image API vs data API)	94×
Person detection — YOLOv7-tiny NPU	~10 ms/frame
System throughput (DET_EVERY=3, 1.5× slowdown)	~5–6 FPS effective
HRNet Golf v2 validation AP (face-on)	0.918
INT8-compatible keypoints	17/17 (100%)
Combined NEF size	70 MB
Pro reference database	18 sequences, 7 professional golfers
DTW comparison time	< 200 ms (40-frame subsampling)

Table 6. System performance summary.

> **Pitfall:** *Code using generic_data_inference appears to work but runs at 6,500 ms/frame. Always use generic_image_inference with BGR565 input. The 94× speedup is not in Kneron documentation — it must be discovered empirically.*

IV-E Known Limitations

- **Throughput:** ~5–6 effective FPS is below real-time video rate. The 1.5× slowdown and DET_EVERY=3 partially compensate. NPU-accelerated post-processing is identified as future work.
- **Monocular 2D:** Single-camera estimation cannot fully resolve 3D body rotation. The L/R lock reduces but cannot eliminate errors during extreme finish-position twist.
- **DTW limitations:** Comparison is position-based — correct positions with wrong timing scores high. Phase-gated DTW (compare within detected phases only) is future work.
- **Domain specificity:** Fine-tuning and L/R fix are specific to face-on golf. The INT8 scaling and calibration findings generalise to other domains and NPU platforms.

V. CONCLUSION

This paper presented a complete methodology for deploying heatmap-based pose estimation on an INT8-only edge NPU, demonstrated through a real-time golf swing analysis system on the Kneron KL730. Three experiments provided quantitative evidence for each design decision. The model selection study showed that architecture determines NPU deployability before accuracy — RTMPose-S, despite AP = 0.971 and strong INT8 signal, is permanently eliminated by a fixed hardware output format constraint. The fine-tuning study quantified that golf domain adaptation incidentally resolves INT8 incompatibility, raising INT8-compatible keypoints from 80.6% to 100%. The calibration study proved that domain-mismatched calibration causes step size errors of 4–13×, directly explaining early deployment failures.

The three core technical findings — the five-criterion selection framework, the fine-tuning to INT8 relationship, and calibration domain sensitivity — are architecture-agnostic and directly applicable to any heatmap pose model deployed on INT8-only edge hardware. The ×1000 scaling technique, dual-model NEF merging, and L/R calibration lock address KL730-specific constraints but represent patterns expected on similar embedded NPU platforms.

The complete web application demonstrates that the system is usable in practice: a golfer can record a swing, receive a DTW-ranked comparison against 18 professional sequences, inspect per-phase joint angle deltas, and receive streaming AI coaching — all running on a single low-power edge board without cloud connectivity. Future work includes 3D pose reconstruction for body twist disambiguation, golf fine-tuning of YOLOv8n-pose to evaluate regression head viability, and a formal user study measuring coaching effectiveness.

REFERENCES

[1] A. Newell, K. Yang, J. Deng, "Stacked hourglass networks for human pose estimation," in *Proc. ECCV*, 2016.

[2] B. Xiao, H. Wu, Y. Wei, "Simple baselines for human pose estimation and tracking," in *Proc. ECCV*, 2018.

[3] K. Sun, B. Xiao, D. Liu, J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proc. CVPR*, 2019.

[4] T. Jiang et al., "RTMPose: Real-time multi-person pose estimation based on SimCC," *arXiv:2303.07399*, 2023.

[5] F. Zhu et al., "Sports pose estimation via transfer learning," in *Proc. ICIP*, 2020.

[6] H. Liu, M. Taniguchi, "Basketball pose estimation via domain adaptation," in *Proc. CVPR Workshops*, 2021.

[7] W. McNally et al., "GolfDB: A video database for golf swing sequencing," in *Proc. CVPR Workshops*, 2019.

[8] S. Srivastava, D. Roy, P. Bhatt, "Athlete motion similarity via dynamic time warping," in *Proc. ICASSP*, 2021.

[9] A. Gholami et al., "A survey of quantization methods for efficient neural network inference," *arXiv:2103.13630*, 2021.

[10] G. Papandreou et al., "Towards accurate multi-person pose estimation in the wild," in *Proc. CVPR*, 2017.